1 Introduction

This project focuses on the design, development, and implementation of an interactive engineering tool that integrates electrical and systems control systems. The objective is to develop a functional and reliable system that allows users to interact with various electrical components and seamlessly import or create one-line diagrams within a specialized component GUI. The software will validate system topology, map zones of protection, and perform critical momentary DC battery load calculations. By identifying worst-case scenarios, the tool provides GFT engineers with actionable insights to ensure maximum substation reliability and safety.

The project provides an opportunity to apply theoretical concepts learned through coursework to the creation of a practical engineering application. Key areas of focus include system design, component integration, programming, testing, and troubleshooting. The completed system is intended to demonstrate the application of engineering methods to solve a real-world problem.

The GUI allows users to construct protection system schematics using a Drag-and-Drop Approach, identifying and visualizing zones of protection, performing momentary current calculations, and accessing a library of electrical components. In addition, the tool provides the ability to import an existing PDF diagram and use it as a background reference while recreating or analyzing the system. These features are intended to make the analysis process more clean, organized, interactive, and easier to understand.

The tool is divided into several primary sections: the Protection Zone Outliner, Momentary Calculation Tool, Component Library, Sandbox, and PDF Importer/Exporter function.

1.1 Project Objectives

The objective of the project is to create an AI-driven tool that will assist in the design process of protection zones within a substation.

The tool will allow the user to: (Needs to be updated for bullet list below later)

Transcribe a One-Line Diagram onto it: Edit the components of the One-Line Diagram Outline the Zones of Protection Calculate the Momentary DC Battery Load Identify the Worst-Case Scenario

2 Tool Overview

2.1 Graphical User Interface

The graphical user interface (GUI) serves as the central workspace for the protection analysis tool. The GUI organizes the different functions into accessible sections, allowing the user to move between diagrams, component selection, protection zone analysis, and electrical calculations without needing to leave the application. The interface is designed around an interactive workspace where components can be placed, connected, and modified as the protection system is developed.

2.2 Component Library

The component library can also be expanded through the Create a component functionality button. This allows a user to create a custom component when a required component is not already included in the library. The ability to create customized components makes the tool more flexible and allows it to accommodate different protection-system configurations and project requirements. The Component Library provides the users with a collection of electrical and protection system components that industries use that can be used to construct one-line diagrams. The Component Library allows for both commonly used and custom created components to be selected and placed into a workspace for quicker access and efficiency, instead of needing the user to manually draw each component into the schematics.

2.3 Protection Zone Outliner

The Zone Outliner will be used to organize and visualize the different areas of Zones of Protection found within the electrical schematics. Within the sandbox, once the components and circuit connections are established, the Zone Outliner would create a display that would identify the protection zones. The purpose of having this feature is make it easier in relationships between the schematic components and their associated protection zones to be easier to understand. Separating the electrical system into individual zones, would allow easier identification among it's users for which components are included within a particular zone and how evaluating the different parts of the system is protected.

2.4 Momentary Calculation

The Momentary Calculation Tool provides a section that is solely focused on performing calculations associated with the momentary electrical load of the protection system for the given One-Line Diagrams. The tool is intended to reduce the need for the user to perform these calculations separately and manually. By providing the calculation functionality within the same GUI used to develop the protection diagram, the user can connect the calculated results to the components and protection zones that are being analyzed in the GUI. This helps maintain a workflow that is concentrated and allows the electrical analysis to be performed alongside the development of the system diagrams in pending projects.

2.5 PDF

After importing the PDF, the diagram can be displayed in the background of the Sandbox while the user recreates with additional component edits/changes or analyzes the system using the available components. The background image as well as the grid can be toggled on or off, allowing the users to switch between viewing the original diagram and viewing of the newly created components within the sandbox.

2.6 Sandbox

The Sandbox is the main workspace where users can create, edit, and analyze their protection system diagrams. The Sandbox is the main workspace within the GUI where users can create, edit, and analyze their protection system diagrams are mainly focused. It allows users to drag and drop components from the Component Library into the workspace and arrange them to represent an electrical protection system. Components can then be connected to one another to create the desired electrical configuration. The sandbox provides a grid layout, allowing for precise component placements, changing of the grid opacity, while the interactive tools would enable the users to connect the components once placed. Users can also save their work, export the diagrams, and share the files with colleagues for any review and feedback that they may have.

The Sandbox is intended to be an interactive environment where the user can experiment with different configurations to be added to the diagram without modifying the original diagram. Components can be repositioned, circuit connections can be adjusted as the system is developed. This enables the Sandbox

as the primary area for creating and modifying the protection system diagram in the GUI.

The sandbox supports real-time updates, so any changes made to the diagram are immediately reflected in the analysis results.

3 Step-by-Step Instructions

4 Engineering Calculation and Validation

5 Testing and Verification

6 Future Improvements

Future development of the engineering tool could bring additional features and improvements to expand its capabilities, more advanced electrical protection analysis of the diagrams. Future development of the software should focus on expanding its engineering capabilities, improving automation, and enhancing the overall user experience. A potential improvement is in the integration of Artificial Intelligence (AI) for the automated diagram recognition. This functionality could allow users to import existing PDF engineering schematics and automatically identify electrical components, symbols, labels, and connections without the additional need to click a verification button. The recognized components could then be converted into editable objects within the Sandbox, reducing the time required to manually recreate existing diagrams.

Additional Future Improvements: Automated Topology Validation - Improve the detection of disconnected components, incorrect connections, and incomplete electrical component configurations.

Improved Analysis Calculation Capabilities - Expand beyond momentary DC battery load calculations to include additional protection and electrical system analyses.

Improved PDF Importing- Enhance background alignment, scaling, symbol recognition, and diagram conversion capabilities.

Expanded Component Library - Include additional substation equipment, protective relays, transformers, circuit breakers, and electrical devices.

Automatic Report Generation - Generate engineering reports containing one-line diagrams, calculation results, protection-zone information, and identified system issues.